ISYE 6740 - Project Proposal

Team #: Group 89
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Project Title: Hospital Bill Predication Using Claims Data

I. Problem Statement

Overview

Hospital cost prediction using claims data is crucial in healthcare management, insurers, and patients. Since the increased financial burden on healthcare providers and patients, hospital cost prediction has become an essential topic in healthcare analytics. It supports institutions in allocating resources, managing risk, and even making policies. Claims data, consisting of patient demographics, diagnosis codes, procedures, and insurance information, is one of the most valuable sources for making such predictions. Also, the rise of machine learning and statistical models could improve the accuracy of such estimation and then propel this field forward. Claims data generally includes patient demographic (age, gender, region), medical history (diagnosis codes, treatment plans, prior hospital visits), procedures performed (procedure codes, surgery type, test results), medication prescribed, length of stay, and insurance plan details (copays, deductibles, insurer-paid portion). Because claims data include patients' private information, they are always hard for the public to obtain.

We explored some major papers that researched claims data. According to **Zhao et al. (2018)**, claims data is used for cost prediction and assessing healthcare utilization and outcomes. These longitudinal data cover a broad patient population, making them suitable for prediction. Early studies predicted hospital costs using traditional statistical models, which focused on the linear relationship between the variables and the cost. In a study by **Zuvekas and Olin (2009)**, linear regression models were used to estimate healthcare expenditures in the U.S., revealing the statistical meaningfulness of this method for initial cost prediction. **Manning and Mullahy (2001)** introduced the use of Generalized Linear Model for healthcare cost data, which often shows overdispersion and non-linearity. They suggested that GLMs perform better than standard linear regression in capturing cost variability across different types of patients.

Nowadays, machine learning methods have been introduced to research. As noted by **Obermeyer et al. (2016)**, Random Forest models have been used to predict healthcare costs and outcomes by aggregating the predictions of multiple decision trees. Random Forests are particularly effective in scenarios with large, complex datasets, such as claims data. A study by **Jin et al. (2020)** demonstrated Gradient Boosting Machines, known for their superior performance in structured data tasks, such as XGBoost and LightGBM, outperformed traditional regression models in inpatient healthcare cost prediction because they capture the non-linearities in claim data.

As healthcare data becomes increasingly complex, integrating new data sources and advancements in machine learning are expected to improve the accuracy and applicability of hospital cost prediction models further. This will ultimately lead to better resource allocation, cost management, and patient triage in the healthcare sector.

Data Source

Our data source for this project is a healthcare dataset from Kaggle (2). This dataset is a synthetic healthcare dataset that was designed to mimic real-world healthcare data. The datasets have 55,500 rows

and 15 columns. Each column provides specific information about the patient, their admission, and the healthcare services provided. Below is a brief explanation of each column:

- **Name:** Name of the patient associated with the healthcare record.
- **Age:** The patient's age at the time of admission is expressed in years.
- **Gender:** Indicates the patient's "Male" or "Female."
- **Blood Type:** Patient's blood type (e.g., "A+", "O-", etc.).
- **Medical Condition:** Primary medical condition or diagnosis associated with the patient
- **Date of Admission:** The date the patient was admitted to the healthcare facility.
- **Doctor:** The name of the doctor responsible for the patient's care during their admission.
- **Hospital:** Healthcare facility or hospital where the patient was admitted.
- **Insurance Provider:** Patient's insurance provider (e.g., "Aetna," "Blue Cross," "Cigna," "UnitedHealthcare," and "Medicare.")
- **Billing Amount:** The amount of money billed for the patient's healthcare
- **Room Number:** The room number where the patient was accommodated during their admission.
- **Admission Type:** Specifies the type of admission, which can be "Emergency," "Elective," or "Urgent," reflecting the circumstances of the admission.
- **Discharge Date:** The date on which the patient was discharged from the healthcare facility
- **Medication:** Identifies a medication prescribed or administered to the patient during their admission. Examples include "Aspirin," "Ibuprofen," "Penicillin," "Paracetamol," and "Lipitor."
- **Test Results:** Medical test conducted ("Normal," "Abnormal," or "Inconclusive")

II. Proposed Methodology

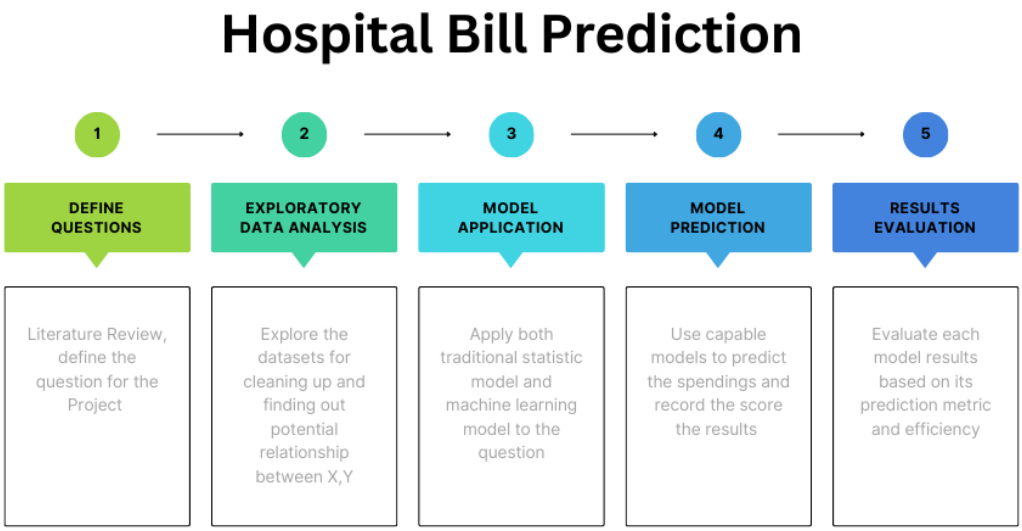

Figure 1: Project Methodology

For our project, we plan to use the healthcare dataset described above to create a predictive model for hospital bills. We have labeled features within the data, so we plan on using a supervised learning methodology to make the prediction model. As shown in Figure 1, we plan to do an exploratory analysis of all the attributes in the dataset. Then, we will apply the proposed machine learning models to the potential relationship we discovered in previous procedures. Further, we will evaluate the results of each model and process prediction using testing datasets. Analyzing the predicted results will be the last step.

Data Preprocessing and Exploratory Analysis

Before prediction, it's essential to clean and prepare the data. We plan and have already started to perform the following:

- **Remove duplicates and outliers:** Claims might have duplicates or data entry errors.
- **Handle missing values:** Impute missing values with averages, medians, or advanced imputation techniques.
- **Categorical data encoding:** Convert categorical features (e.g., diagnosis, procedures) into numerical formats using one-hot or label encoding.
- **Standardize/normalize numerical features:** Scale numerical data (e.g., age, cost) for better model performance.
- **Feature engineering:** Derive new features such as age groups, chronic illness flags, or the number of procedures per patient.

To perform feature selection, we must convert categorical values into numerical representations. As part of our data preprocessing and exploratory data analysis (EDA) process, we will use a combination of three standard encoding techniques: Label Encoding, Ordinal Encoding, and One-Hot Encoding.

Label encoding assigns a unique integer value to each category in a categorical variable, while ordinal encoding assigns an integer value based on the data order. One-hot encoding converts each category into a binary vector, where only one bit is 1 (hot) and the rest are 0 (cold).

Feature Selection

Due to our dataset's large number of categorical features, we can leverage Pearson's Chi-Squared Test for feature selection. The Chi-Squared test is a statistical test applied to categorical data to evaluate the likelihood that any observed difference between the sets arose by chance. Since not all features equally contribute to cost prediction, below are some additional feature selection techniques we could leverage:

- **Correlation matrix:** Helps to remove highly correlated features.
- **Feature importance from tree-based models:** Helps prioritize features that contribute most to cost prediction.
- **Principal Component Analysis (PCA):** Reduces dimensionality by identifying uncorrelated feature combinations.
- **Mutual Information:** A metric used to measure the amount of information that one feature contributes to predicting the target variable.

Model Prediction Approach

There are several approaches for hospital cost prediction using claims data. Below are the standard methods:

Regression-Based Methods	Tree-Based Models	Neural Networks
Regression models predict continuous outcomes like hospital costs based on input features. • **Linear Regression:** A basic model that assumes a linear relationship	Tree-based models capture non-linear relationships and are more flexible than linear regression. • **Random Forest:** A collection of decision trees that predict hospital costs by	Powerful for capturing complex relationships and may require more computational resources. • **Feedforward Neural Networks (FNN):** FNNs can model non-linear relationships

between features (e.g., age, length of stay) and hospital costs. • **Lasso and Ridge Regression:** These are forms of regularized regression that help prevent overfitting by penalizing large coefficients. • **Generalized Linear Models (GLM):** GLMs (e.g., Poisson or Gamma regression) are often used in insurance for cost prediction, especially when the data has non-normal distributions.	averaging the results of many trees, reducing overfitting. • **Gradient Boosting (e.g., XGBoost, LightGBM):** A powerful method that builds trees sequentially to correct previous mistakes, often outperforming Random Forest in structured data.	between features and hospital costs but require extensive training data. • **Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM):** For predicting costs based on time-series data, like hospitalizations over time, RNNs or LSTMs could be useful.

Figure 2: Predication Model Methods

After the data is cleansed and feature engineering is processed, we plan on splitting the data for training and testing. We will then start testing the various prediction approaches mentioned above and evaluating initial accuracy scores.

III. Model Evaluation Strategies

Lastly, evaluating our model plays a pivotal role in judging the overall performance of the model. There are many critical metrics for assessing the predictive models, below are some of the common techniques we plan to focus on:

- **Mean Absolute Error (MAE):** The average absolute difference between predicted and actual costs.
- **Mean Squared Error (MSE) or Root Mean Squared Error (RMSE):** Square of the error to penalize larger mistakes.
- **R-squared:** The proportion of variance in hospital costs explained by the model.
- **Precision/Recall**: Precision is the ratio of correctly predicted positive observations to all predicted positives and gauges how well the model forecasts the positive outcomes. Recall is the ratio of correctly predicted positive observations to all actual positive observations and gauges how well the model can capture each pertinent instance.
- **F1 Score:** Metric that measures a model's accuracy by combining the precision and recall scores of a model.

Leveraging these metrics will allow us to pinpoint areas for model improvement and ensure that our final model delivers reliable predictions in real-world hospital settings.

References

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Appendix

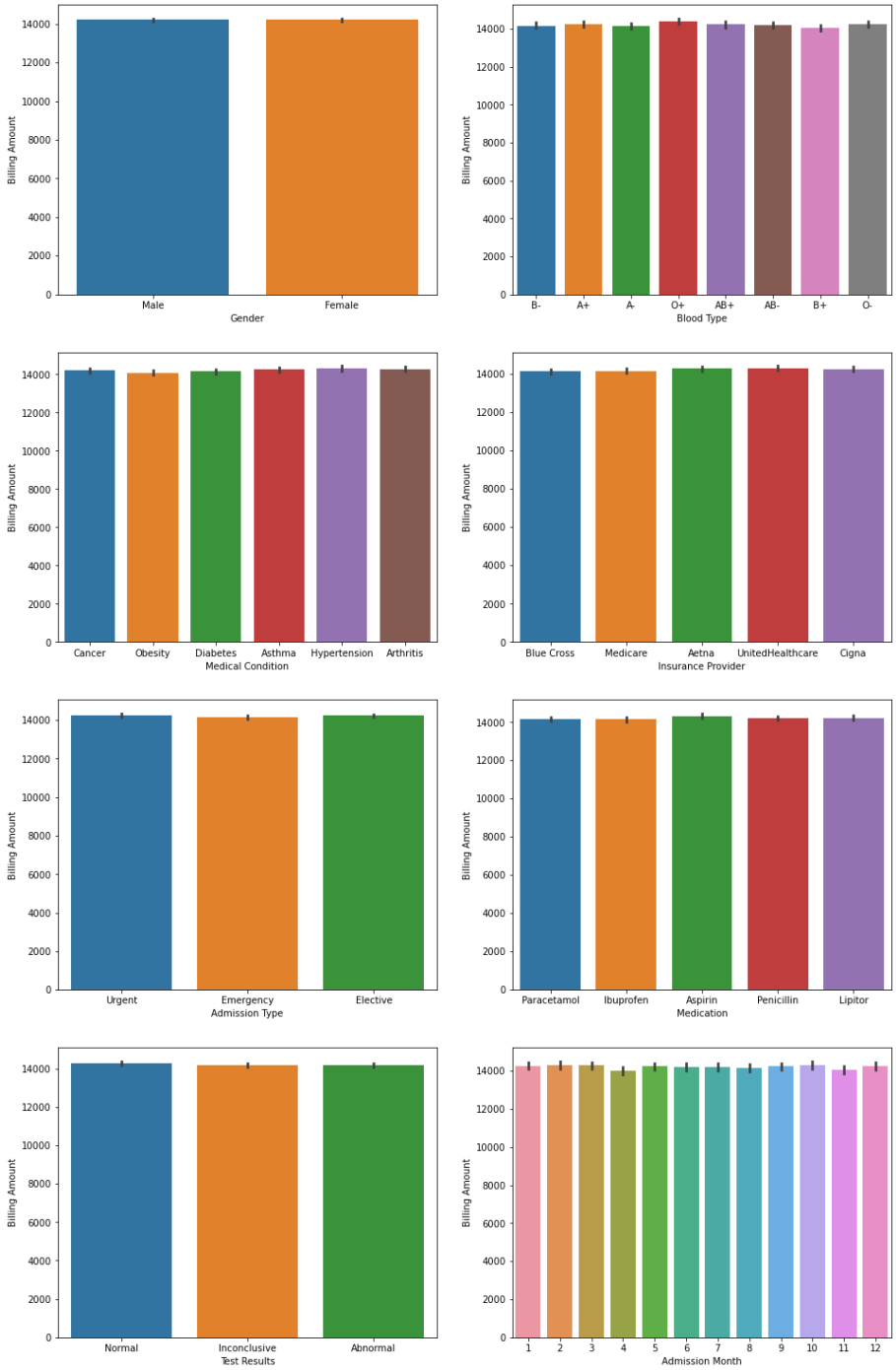

Figure 3: Exploratory Analysis of Categorical Variables and Billing Amount